

# PROJECT DESIGN PHASE-II

## Solution Requirements Specification (SRS)

|         |                                               |               |                          |
|---------|-----------------------------------------------|---------------|--------------------------|
| Date    | 18 June 2026                                  | Project Name  | ShopEz (MERN Full Stack) |
| Team ID | --                                            | Maximum Marks | 4 Marks                  |
| Student | SIDDANI SRI CHARAN (charaniamgreat@gmail.com) |               |                          |

### Functional Requirements (FR)

| ID    | Requirement (Epic)                                  | Description & Logic                                                                       |
|-------|-----------------------------------------------------|-------------------------------------------------------------------------------------------|
| FR-1  | User Registration<br>• Form Input • Validation      | Mechanism for new users to create accounts using email and secure password hashing.       |
| FR-2  | User Authentication<br>• JWT Session • Login/Logout | Secure login system providing stateless authentication via JSON Web Tokens.               |
| FR-3  | Product Catalog<br>• Dynamic Fetch • Grid View      | Displaying Men, Women, and Kids categories with real-time data from MongoDB.              |
| FR-4  | Advanced Search<br>• Regex Matching                 | Allowing users to query the product database using keywords with instant results.         |
| FR-5  | Interactive Filtering<br>• Price • Category • Size  | Dynamic filtering of product lists without page reloads using React state management.     |
| FR-6  | Cart Management<br>• Persistence • CRUD             | Add, remove, and update quantities of products in a persistent shopping cart.             |
| FR-7  | Secure Checkout<br>• Address • Payment Mock         | Process for capturing shipping details and reviewing order totals before final placement. |
| FR-8  | Order Tracking<br>• Status Updates                  | Interface for users to view current status (Pending/Shipped/Delivered) of their orders.   |
| FR-9  | Admin Product CRUD<br>• Image Upload • Pricing      | Restricted dashboard for adding, editing, and deleting store inventory items.             |
| FR-10 | Admin Order Flow<br>• Management • Statistics       | Centralized control for admins to manage global orders and monitor sales trends.          |

### Non-Functional Requirements (NFR)

| ID    | Attribute      | Requirement Specification                                                     | Importance |
|-------|----------------|-------------------------------------------------------------------------------|------------|
| NFR-1 | Security       | Passwords must be encrypted via Bcrypt; API endpoints protected by JWT.       | CRITICAL   |
| NFR-2 | Performance    | Initial page load < 2s; API response time for product queries < 500ms.        | HIGH       |
| NFR-3 | Scalability    | Architecture must handle horizontal scaling using MongoDB Atlas partitioning. | HIGH       |
| NFR-4 | Responsiveness |                                                                               | HIGH       |

| ID    | Attribute      | Requirement Specification                                                        | Importance |
|-------|----------------|----------------------------------------------------------------------------------|------------|
|       |                | Mobile-first adaptive design for seamless usage across smartphones and desktops. |            |
| NFR-5 | Availability   | System uptime should target 99.9% availability during peak shopping hours.       | CRITICAL   |
| NFR-6 | Data Integrity | Referential integrity maintained between Users, Products, and Order collections. | CRITICAL   |